Blocaïne

Installation and Evaluation Procedure

(The open-source solution for automation)

Table of contents

1 General information

1.1 Evaluation setup

To evaluate **Blocaïne**, you only need a single PC on which both the **Bloc Editor** and the **Interpreter** will be installed.

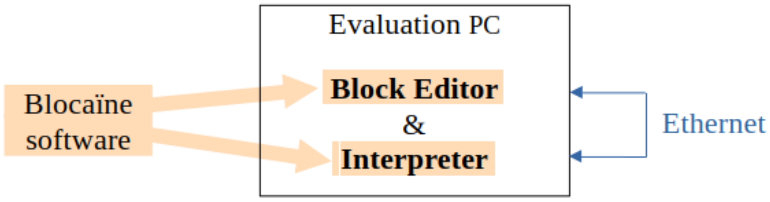


Figure 1a : Evaluation Architecture

1.2 Standard setup

For a real automation project, the **Block Editor** must be installed on a development PC and the **Interpreter** on a Target computer.

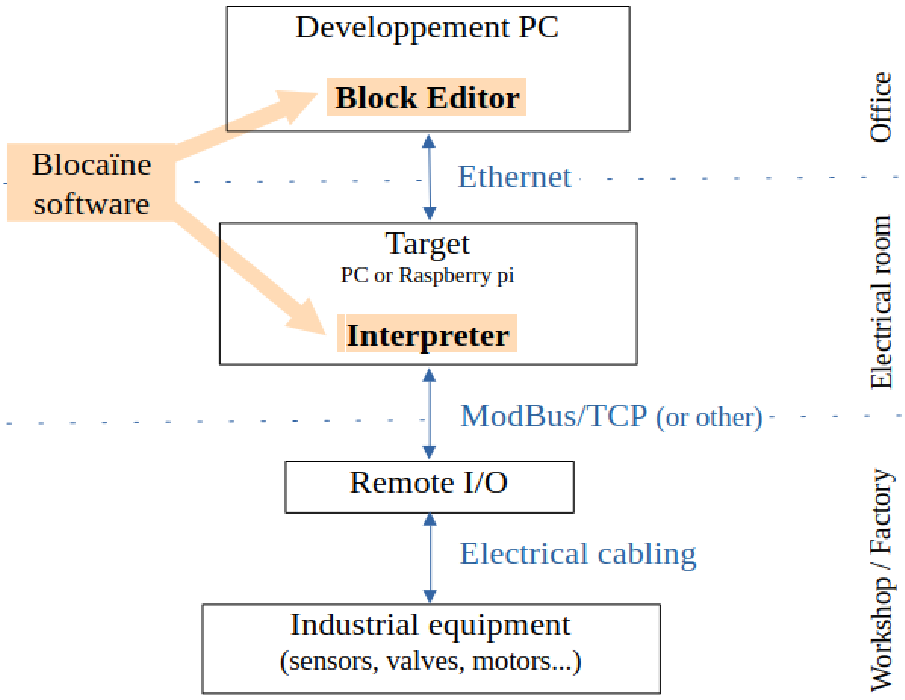


Figure 1 : General Architecture

2 Requirements

To be compatible with **Blocaïne**, a computer must have one of the following operating systems: Windows, Linux, or macOS, at least 100 MB of free disk space, and Python version 3.6 or later.

3 Installing Blocaïne

3.1 Installation procedure for evaluation

Installation takes about 3 minutes. It is simplified because the downloadable version is preconfigured and ready to use:

1. No network configuration is required, since the **Interpreter** and the **Block Editor** run on the same PC.
2. No Python modules need to be installed, because the OPC, Modbus, and GPIO features that require specific Python modules are disabled by default.

To install the “evaluation” version of Blocaïne, follow these steps:

1. Download **Blocaïne** (see chapter 4)
2. Launch the **Interpreter** (see chapter 7)
3. Launch the **Block Editor** (see chapter 8)
4. Verify that **Blocaïne** is working correctly (see chapter 9)

3.2 Installation procedure for a real automation project

This procedure explains how to turn the “evaluation” version into a fully usable version:

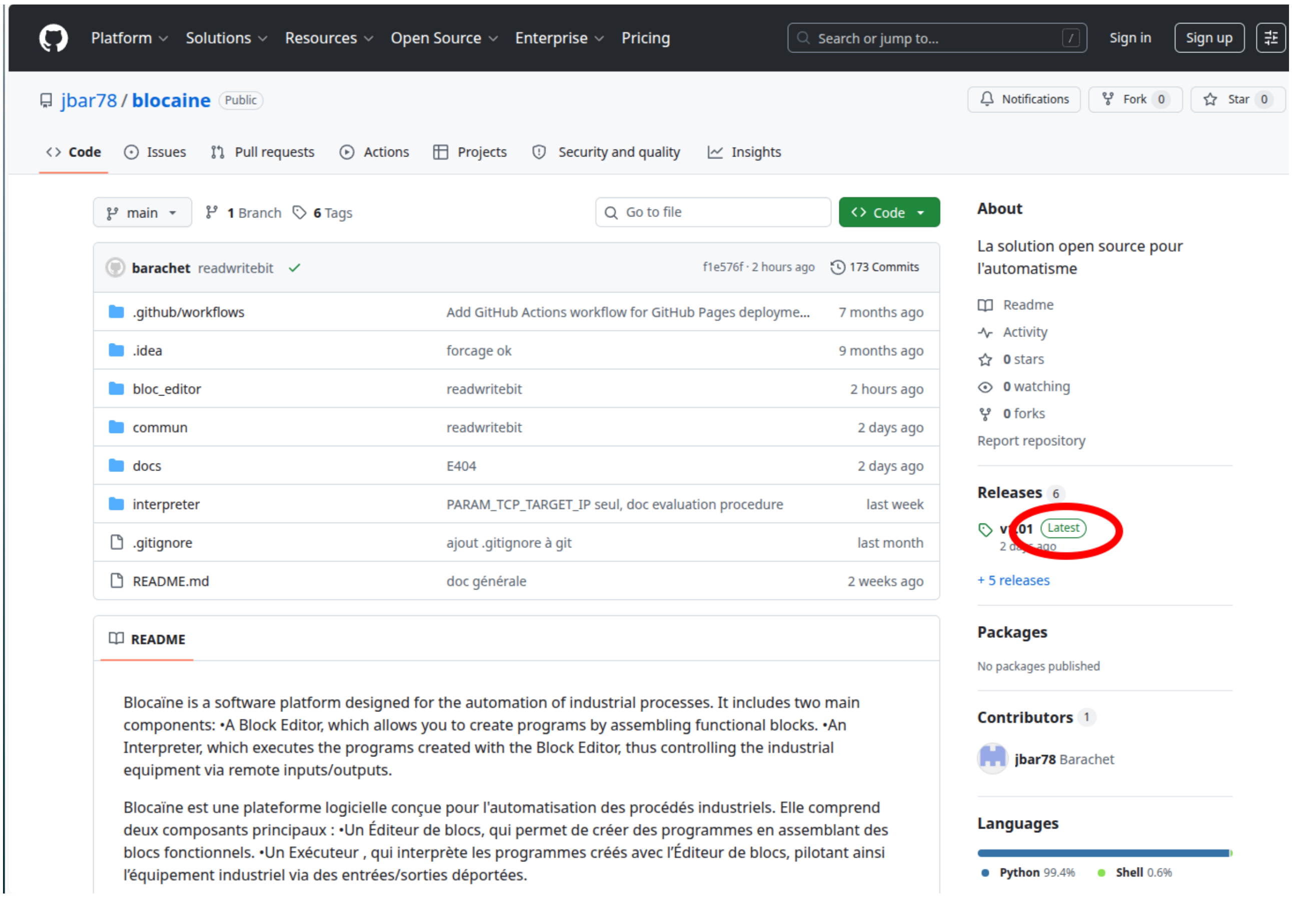
1. Install **Blocaïne** on your development PC by following the evaluation installation procedure above.
2. Install **Blocaïne** on Target machine (see chapter 5)
3. Configure the network (see chapter 6)
4. Launch the **Interpreter** on Target machine (see chapter 7)
5. Launch the **block Editor** on the development PC (see chapter 8)
6. Verify that **Blocaïne** is working correctly (see chapter 9)

4 Downloading Blocaïne

Using your preferred web browser, go to:

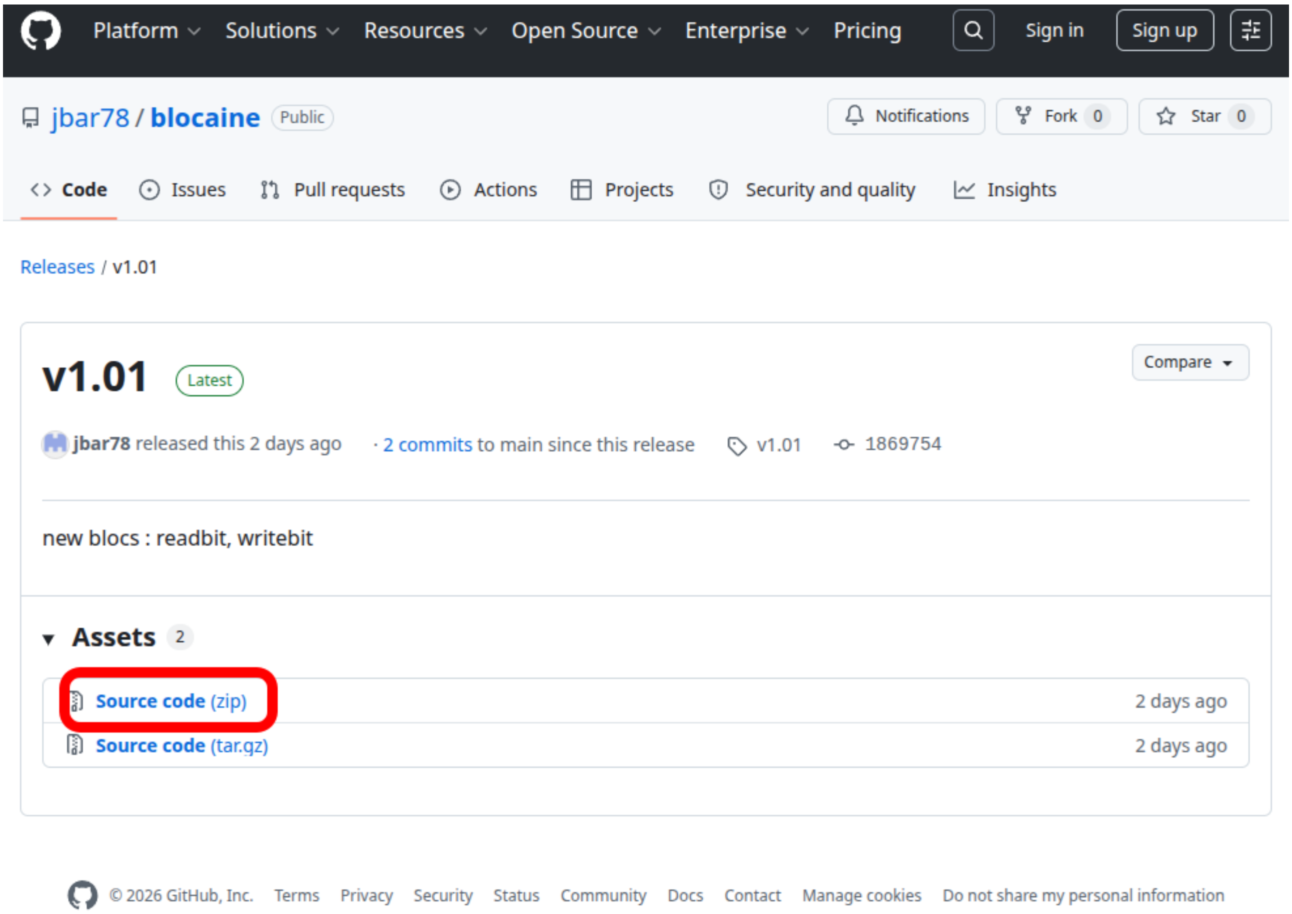
https://github.com/jbar78/blocaine.git

You will arrive at this page:



Click “Latest” on the right.

You will be taken to the latest release page:



Click "Source code (zip)", choose where you want to save the ZIP file, and once downloaded, **extract** it. You should then obtain a new directory named «blocaine-*x.y*» (*x.y* being the version)..

Great, the installation is already finished.

5 Installing Blocaïne on a Target machine

If you only want to use the “evaluation” version of **Blocaïne**, this chapter does not apply. However, if you want to convert the “evaluation” version into a fully usable version for an automation project, you must install the **Interpreter** on a target machine.

If your target machine has internet access, you can download Blocaïne again on the target machine. You can also copy the `blocaine-x.y` directory from the development PC to the target machine.

5.1 Using ModBus Remotes I/O

If you want the **Interpreter** to manage Modbus inputs and outputs, you must enable this feature by editing the `PARAM_CONFIG.py` file located in the « interpreter » directory on the target machine and changing `PARAM_CONFIG_MODULE_MODBUS` from `False` to `True`. You must also install the Python module **`pymodbus`** on the target machine.

5.2 Using OPC protocole

If you want the **Interpreter** to include an OPC UA server so that it can connect to an HMI or SCADA system, you must enable this feature by editing the `PARAM_CONFIG.py` file located in the « interpreter » directory on the target machine and changing `PARAM_CONFIG_MODULE_OPC` from `False` to `True`. You must also install the Python module **`opcua`** on the target machine.

5.3 Using Raspberry pi GPIO pins

If the target machine is a Raspberry Pi and you want the **Interpreter** to control the GPIO pins, you must enable this feature by editing the `PARAM_CONFIG.py` file located in the « interpreter » directory on the target machine and changing `PARAM_CONFIG_MODULE_GPIOZERO` from `False` to `True`. You must also install the Python module **`gpiozero`** on the target machine..

5.4 Installing required Python module

5.4.1 On Raspberry pi

If your target machine is a Raspberry Pi running Pi OS, it is recommended to install new Python modules in a virtual environment. Proceed as follows:

Opend a terminal and type:

```
sudo apt update
sudo apt install python3 python3-pip python3-venv
```

Go to the «blocaine-*x.y*/interpreter» directory :

```
python3 -m venv --system-site-packages venv-blocaine # create an environment
source venv-blocaine/bin/activate # activate the environment
pip install pymodbus # install the module in the environment
pip install opcua # install the module in the environment
pip install gpiozero # install the module in the environment
```

5.4.2 On Linux (Debian, Fedora, Ubuntu...)

If your target machine is a Linux PC, you can install new Python modules in a virtual environment in the same way as on a Raspberry Pi, except that you should not install the **`gpiozero`** Module.

5.4.3 On Windows

It is not ideal to use a Windows PC as the target, because Windows is not known for handling real-time constraints properly, but it is possible. You can install new Python modules in a virtual environment, except that you should not install the **`gpiozero`** Module.

5.4.4 Python 3 virtuel environment

For the **Interpreter** to use the previously created « venv-blocaine » virtual environment, activate it from the « interpreter » directory with the following command:

```
Source venv-blocaine/bin/activate
```

Before launching the **Interpreter** with:

```
python3 main.py
```

Make sure that the `launch_interpreter.sh` file contains these two commands.

6 Network Configuration

If you only want to use the “evaluation” version of **Blocaïne**, this chapter does not apply. However, if you want to convert the “evaluation” version into a fully usable version for an automation project, you must configure an Ethernet network, wired or Wi-Fi, between the development PC and the target machine.

The address of the target machine is defined by the `PARAM_TCP_TARGET_IP` parameter in the `PARAM_TARGET_ADDRESS.py` file located in the « bloc_editor » directory on the development PC. The default value is `"127.0.0.1"` and must be replaced with the target machine address on your network.

7 Launching the Interpreter

7.1 Under Windows

To launch the **Interpreter**, open the « blocaine-x.y\interpreter »directory in File Explorer, then double-click `main.py` to start it. If the firewall detects that the **Interpreter** is opening new port, click Allow.

7.2 Under pi OS (Raspberry pi)

To launch the **Interpreter**, open the « blocaine-x.y/interpreter » directory in the file manager, then left-click `launch_interpreter.sh`, then left-click “Run in Terminal”.

7.3 Under Linux

To launch the Block Editor, open the « blocaine-*x.y*/interpreter » directory in the file manager, right-click `launcher_bloc_editor.sh`, then left-click “Run as a program”.

7.4 Under MacOS

Lancez le fichier « main.py » se trouvant dans le répertoire « blocaine-*x.y*/interpreter » avec l’interpréteur Python3.

7.5 Verify that the Interpreter is Operational

When the **Interpreter** terminal displays:

Target is ready 😊

This indicates that the Executor is operational; you can now launch your browser and open the page http://*TargetIP@*:8080 (port 8080) to navigate the menu provided by the **Interpreter**.

TargetIP@ is the IP address of the target machine defined by the PARAM_TCP_TARGET_IP parameter in the "PARAM_TARGET_ADDRESS.py" file, located in the "bloc_editor" directory. If you are using the "evaluation version," this parameter is set to "127.0.0.1"; otherwise, it is the target address you defined on the development PC.

8 Launching the block Editor

8.1 Under Windows

Open the “blocaine-*x.y*\bloc_editor” directory, then double-click “main.py.”

8.2 Under Linux

Open the “blocaine-*x.y*/bloc_editor” directory in your file explorer, right-click on “launch_bloc_editor.sh," then select “Run as a program.”

8.3 Under MacOS

Open the “blocaine-*x.y*/bloc_editor” directory, the run “main.py” in the terminal using Python3

The Python interpreter will start, and both a terminal and the **Block Editor**’s graphical interface will appear. The **Block Editor** automatically connects to the Blocaïne **interpreter**.

You can now create your first **user** blocks, upload them, and execute them in the **interpreter.**

8.4 Verify that the Block Editor is Operational

When the **Block Editor** is launched, a terminal and the **Block Editor**’s graphical interface will appear. The **Block Editor** automatically connects to the Blocaïne **interpreter**.

You can now create your first user blocks, upload them, and execute them in the interpreter.

9 Verify that Blocaïne is working properly

9.1 Checking the connection

To check whether the **interpreter** is properly connected to the **Block Editor**, look at the status bar located at the bottom of the **Block Editor**'s graphical interface. It should display the message "Target: Connected"

9.2 Editing the user block <loop>

Using the **Block Editor**, edit the user block <loop> via the "Bloc" menu. Then click "Open," select the file "loop.bloc," and click "Open." The block should appear in the graphical area.

9.3 Compilation, download & execution

To compile the user block <loop>, download it to the Executor, and run it, use the "Target" menu. Then click "Check, Transfer & Execute <loop>" or click the ⚡ icon.
After compilation, a popup will ask you to confirm whether you want to download and run the <loop> block. Click "Yes."
You can verify that the <loop> block has started correctly by using the web server (http://127.0.0.1:8080).

9.4 Debugging

To view the current values of the variables in the <loop> block, enable "Monitoring" mode via the "Target" menu. Then click "Monitoring (Start/Stop) [Space]," or press the "Space" key, or click the 👁️ icon.

10 Désinstallation de **Blocaïne**

Étant donné que l'installation consiste simplement au téléchargement d'un fichier zip puis à son dézippage, que si vous avez installé des modules Python, vous l'avez fait dans un environnement virtuel, pour désinstaller **Blocaïne** il suffit de supprimer le fichier zip ainsi que le répertoire dézippé.